FitCheck — Relational Database Schema & Constraint Map

3NF Normalized Relational Design | Primary Keys, Foreign Keys, Referential Integrity, and Domain Constraints

InnoDB Engine

USER

PK | user_id : INT AUTO_INCREMENT

name : VARCHAR(100) NOT NULL

UQ | email : VARCHAR(255) NOT NULL

password_hash : VARCHAR(255) NOT NULL

CK | role : VARCHAR(10) IN ('USER','ADMIN')

created_at : TIMESTAMP DEFAULT CURRENT_TIMESTAMP

Referenced by: BODY_PROFILE, PURCHASE, FIT_REVIEW

BODY_PROFILE

PK | profile_id : INT AUTO_INCREMENT

FK, UQ | user_id : INT -> USER(user_id)
[ON DELETE CASCADE, ON UPDATE CASCADE]

CK | height : DECIMAL(5,2) > 0

CK | waist : DECIMAL(5,2) > 0

CK | hip : DECIMAL(5,2) > 0

CK | thigh, inseam : DECIMAL(5,2) > 0

CK | preferred_fit : IN ('TIGHT','REGULAR','RELAXED')

updated_at : TIMESTAMP ON UPDATE CURRENT_TIMESTAMP

Trigger: trg_before_body_profile_update

BRAND

PK | brand_id : INT AUTO_INCREMENT

UQ | brand_name : VARCHAR(100) NOT NULL

CK | sizing_system : IN ('ALPHA','NUMERIC','WAIST_BASED')

Referenced by: GARMENT (RESTRICT on delete)

GARMENT

PK | garment_id : INT AUTO_INCREMENT

FK | brand_id : INT -> BRAND(brand_id)
[ON DELETE RESTRICT, ON UPDATE CASCADE]

garment_name : VARCHAR(150) NOT NULL

CK | garment_category : IN ('Jeans','Trousers')

CK | fit_type : IN ('STRAIGHT','SLIM','SKINNY',...)

description : TEXT NULL

Index: idx_garment_brand

PURCHASE

PK | purchase_id : INT AUTO_INCREMENT

FK | user_id : INT -> USER(user_id) [RESTRICT]

FK | garment_size_id : INT -> GARMENT_SIZE [RESTRICT]

purchase_date : DATE DEFAULT (CURRENT_DATE)

CK | outcome : IN ('KEPT', 'RETURNED', 'EXCHANGED')

CK | fit_satisfaction : IN ('GOOD', 'AVERAGE', 'POOR')

CK | fit_preference : IN ('TIGHT', 'REGULAR', 'RELAXED')

Calibrated Fit Evidence: outcome='KEPT' AND fit_satisfaction='GOOD'

Indexes: idx_purchase_user, idx_purchase_size

FIT_REVIEW

PK | review_id : INT AUTO_INCREMENT

FK | user_id : INT -> USER(user_id) [RESTRICT]

FK | garment_size_id : INT -> GARMENT_SIZE [RESTRICT]

CK | overall_fit : IN ('TOO_SMALL','SLIGHTLY_SMALL','TRUE_TO_SIZE',...)

CK | waist_fit, hip_fit, thigh_fit : IN ('TIGHT','GOOD','LOOSE')

CK | length_fit : IN ('SHORT','GOOD','LONG')

CK | review_text : TEXT [CHAR_LENGTH(TRIM(review_text)) > 0]

created_at : TIMESTAMP DEFAULT CURRENT_TIMESTAMP

Audit Triggers: trg_after_review_insert / update / delete -> REVIEW_AUDIT

Vector Search Sync: Redis Hash 'review_vector:{id}' (384d embedding)

REVIEW_AUDIT (Audit Logging)

PK | audit_id : INT AUTO_INCREMENT

review_id : INT (Target review reference)

action : VARCHAR(10) ('INSERT', 'UPDATE', 'DELETE')

user_id : INT NULL | changed_at : TIMESTAMP

details : TEXT

GARMENT_SIZE

PK | garment_size_id : INT AUTO_INCREMENT

FK | garment_id : INT -> GARMENT(garment_id)
[ON DELETE RESTRICT, ON UPDATE CASCADE]

size_label : VARCHAR(20) NOT NULL

UQ | uq_garment_size (garment_id, size_label)

Index: idx_garment_size_garment

GARMENT_MEASUREMENT

PK | measurement_id : INT AUTO_INCREMENT

FK | garment_size_id : INT -> GARMENT_SIZE
[ON DELETE CASCADE, ON UPDATE CASCADE]

CK | measurement_type : IN ('waist','hip','thigh','inseam','rise')

CK | measurement_value : DECIMAL(6,2) > 0

CK | unit : VARCHAR(5) IN ('cm','in') DEFAULT 'cm'

UQ | uq_size_measurement (garment_size_id, type)

Function: fn_convert_to_cm

Referential Integrity & Normalization Verification (3NF Compliance):

1. **Boyce-Codd / 3NF Normalization:** Every non-key attribute is non-transitively dependent strictly on the primary key. Sizing conventions belong to BRAND; style names belong to GARMENT; individual sizing belongs to GARMENT_SIZE; dimensions belong to GARMENT_MEASURE
2. **Referential Cascade vs Restrict Rules:** Dependent child entities (BODY_PROFILE, GARMENT_MEASUREMENT) use ON DELETE CASCADE to prevent orphaned records. Critical business records (BRAND, GARMENT, GARMENT_SIZE) enforce ON DELETE RESTRICT to gu
3. **Candidate Key Protection:** Composite unique keys (garment_id, size_label) and (garment_size_id, measurement_type) eliminate duplicate sizes and dimensional conflicts at the database engine level.
4. **Domain Constraints & Enums:** All measurements enforce strictly positive values (> 0); review ratings and purchase outcomes are constrained by explicit CHECK constraints.